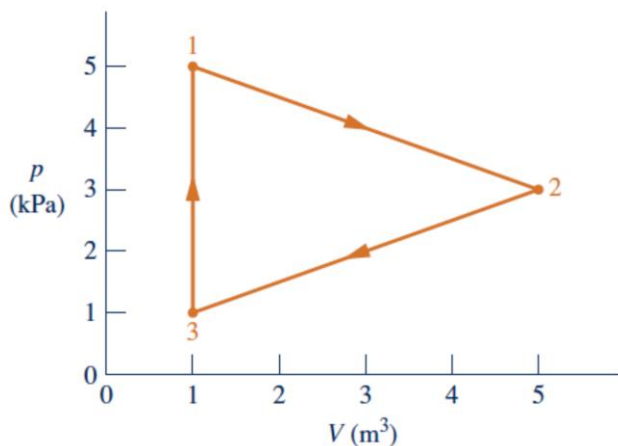


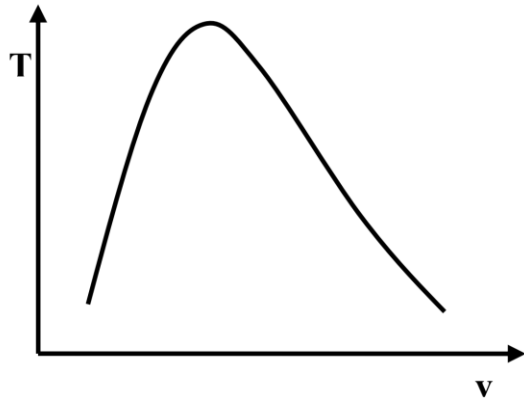
Homework 2 – Due February 3 at 11:59 PM

Please submit a pdf of completed HW through Canvas with all problems in a single file. For scanning handwritten work for submission, I've personally found the Genius Scan app works well. You must show your work to earn full credit. Late work will not be accepted.

1. The figure below shows a power cycle executed by a gas in a piston-cylinder assembly. For process 1-2, $U_2 - U_1 = 15 \text{ kJ}$. For process 3-1, $Q_{31} = 10 \text{ kJ}$. Neglecting changes in kinetic and potential energy, determine the following:
 - a. The work for each process. (*Answer check: $W_{12} = 16 \text{ kJ}$, $W_{23} = -8 \text{ kJ}$, $W_{31} = 0$*)
 - b. The heat transfer for process 1-2. (*Answer check: $Q_{12} = 31 \text{ kJ}$*)
 - c. The heat transfer for process 2-3. (Hint: you will need to apply the first law of thermodynamics for both the cycle as a whole and the first law for each individual cycle. Remember that energy is a property and does not change after a complete cycle, the same way that ΔV will be 0 for the cycle.) (*Answer check: $Q_{23} = -33 \text{ kJ}$*)



2. Sketch a T-v diagram similar to that shown below. Sketch a constant pressure line which passes through the liquid-vapor mixture region, and on that line, label the following points/regions:
 - a. Saturated liquid
 - b. Saturated vapor
 - c. Superheated vapor region
 - d. Compressed/subcooled liquid region
 - e. Liquid-vapor mixture region
 - f. Critical point



3. Answer the following questions in a sentence or two:
- Why do cans or bottles of beer and soda sometimes explode if you leave them in the freezer?
 - Why should you never open the radiator cap on a car when the car is hot?
 - What makes “dry ice” (frozen CO_2) “dry?”
 - Why does it take longer to cook food by boiling (e.g. boiling potatoes or pasta) at high elevation when compared to cooking at sea level?
4. For H_2O , determine the specified property at the indicated state.
- Given: $P = 300 \text{ kPa}$, $v = 0.5 \text{ m}^3/\text{kg}$. Find T (Answer check: $T = 133.6 \text{ }^\circ\text{C}$)
 - Given: $P = 150 \text{ bar}$, $T = 220 \text{ }^\circ\text{C}$. Find v (Answer check: $v = 1.1748 \times 10^{-3} \text{ m}^3/\text{kg}$)
 - Given: $P = 1 \text{ MPa}$, $T = 400 \text{ }^\circ\text{C}$. Find v (Answer check: $v = 0.3066 \text{ m}^3/\text{kg}$)
 - Given: $T = 100 \text{ }^\circ\text{C}$, $x = 60\%$. Find v (Answer check: $v = 1.004 \text{ m}^3/\text{kg}$)
5. Fill in the table below for H_2O :

T ($^\circ\text{C}$)	P (bar)	u (kJ/kg)	Phase
	4	1825	
220			Saturated vapor
100	25		
	40	3099.5	